AI-Driven Inverse Design and In-Silico Screening of High-Entropy Metal–Ceramic Gradient Architectures (HE-MCGA) for Engine-Grade Thermo-Mechanical Performance

Executive Summary

Next-generation aerospace propulsion systems such as the LEAP and RISE engine programs of Safran require materials capable of operating beyond the thermal and mechanical limits of conventional superalloys and thermal barrier coatings. Existing metal–ceramic systems suffer from thermal expansion mismatch, interface degradation, oxidation, and limited thermal fatigue resistance under prolonged exposure to temperatures exceeding 1100–1300 °C.

This proposal introduces an AI-driven startup platform for the inverse design and in-silico optimisation of High-Entropy Metal–Ceramic Gradient Architectures (HE-MCGA) tailored for turbine hot-section applications. The platform combines:

- High-entropy refractory alloys (HEAs)
- High-entropy oxides/carbides (HEOs/HECs)
- Functionally graded metal–ceramic architectures
- Physics-informed machine learning
- Generative inverse design
- Multi-objective Bayesian optimisation
- CALPHAD thermodynamics
- Thermo-mechanical finite element simulations

Unlike conventional sequential material development approaches, the HE-MCGA platform simultaneously optimises:

- Alloy composition
- Ceramic phase chemistry
- Gradient architecture
- Interface transition zone
- Thermo-mechanical performance

inside a single AI-driven optimisation loop.

The proposed platform functions as an upstream AI materials intelligence engine that rapidly identifies fabrication-ready architectures for aerospace environments involving:

- Extreme thermal gradients

- Oxidising atmospheres
- Thermal fatigue cycling
- High creep loading
- Residual stress accumulation

Problem Statement

Current aerospace hot-section materials are fundamentally limited:

Existing System	Key Limitation
Nickel superalloys	Temperature ceiling
Thermal barrier coatings	Delamination/spallation
Ceramic matrix composites	Brittleness/interface failure
Conventional metal–ceramic joints	CTE mismatch

Functionally graded hybrid architectures provide a transformative solution by gradually transitioning material behaviour from metallic to ceramic across the component depth. However, the compositional design space is extraordinarily large and computationally intractable using conventional methods alone.

AI-driven inverse design is therefore essential for accelerated discovery of aerospace-grade hybrid architectures.

Proposed Material Systems

The HE-MCGA platform will explore both:

1. High-Entropy Alloy (HEA) Metallic Systems

Primary Refractory HEAs

- Mo–Nb–Ta–W–Cr
- Nb–Ta–Ti–Zr–Hf
- Cr–Mo–V–W–Ti

Functions

- High-temperature strength
- Creep resistance
- Fatigue resistance
- Structural load bearing

2. Ceramic Systems

Phase 1: Industrially Feasible POC Ceramics

- ZrO_2–HfO_2 gradient ceramics
- YSZ-based systems
- Al_2O_3 protective phases

These systems provide:

- Lower technical risk
- Easier manufacturability
- Strong aerospace relevance
- Direct compatibility with current turbine systems

Phase 2: Advanced High-Entropy Ceramic (HEC/HEO) Systems

High-Entropy Oxides (HEOs): $(ZrHfTaNbTi)O_x$

High-Entropy Carbides (HECs): $(HfTaZrNbTi)C$

High-Entropy Borides: $(ZrHfTaNbTi)B_2$

These entropy-stabilized ceramic systems provide:

- Ultra-high temperature capability
- Low thermal conductivity
- Enhanced oxidation resistance
- Improved phase stability
- Strong AI-driven discovery potential

Hybrid HEA–HEC Gradient Architecture Concept

The proposed architecture consists of:

Region	Function
HEA-rich metallic root	Structural strength and toughness
AI-designed transition gradient	Residual stress minimisation
HEO/HEC-rich surface	Thermal and oxidation protection

The AI platform optimises the entire hybrid architecture as a unified thermo-mechanical system rather than treating metallic and ceramic regions independently.

Target AI Output Properties

The AI platform will identify architectures simultaneously achieving:

Target Property	Purpose
High strength at 1000–1200 °C	Structural integrity
Low thermal conductivity	Thermal insulation
Low oxidation rate	High-temperature durability
Reduced CTE mismatch	Stress minimisation
High thermal fatigue resistance	Long service lifetime
Lightweight design	Aerospace efficiency
Manufacturability	AM/powder metallurgy compatibility

AI Methodology

The HE-MCGA platform integrates four core computational modules:

Module	Function
M1	Data curation & materials intelligence
M2	AI property prediction
M3	Inverse design & optimisation
M4	In-silico validation

Core AI Technologies

Generative Inverse Design

Conditional Variational Autoencoders (cVAEs) generate novel compositions directly from target performance requirements.

Physics-Informed Neural Networks (PINNs)

PINNs embed governing thermo-mechanical equations into neural network training for physically consistent predictions across unseen gradient architectures.

Bayesian Optimisation

Multi-objective Bayesian optimisation identifies Pareto-optimal architectures balancing conflicting objectives such as:

- thermal resistance,
- toughness,
- oxidation resistance,
- and density.

In-Silico Validation Pipeline

All shortlisted candidates undergo:

- CALPHAD thermodynamic screening
- Thermal cycling FEA
- Oxidation kinetics modelling
- Residual stress analysis
- Delamination risk prediction

Validation conditions are aligned with LEAP engine turbine operating environments (>1100 °C thermal cycling).

Proof of Concept (POC)

Initial POC System

Metallic Phase

Mo–Nb–Ta–W–Cr refractory HEA

Ceramic Phase

ZrO_2–HfO_2 gradient ceramic

Advanced AI Exploration

Extension toward: $MoNbTaWCr \rightarrow (ZrHfTaNbTi)O_x$ entropy-stabilized hybrid architectures.

POC Deliverables (6–9 Months)

- AI-trained thermo-mechanical surrogate models
- Pareto-optimised gradient architecture database
- CALPHAD stability validation
- Thermal cycling FEA analysis
- Top candidate ranking
- Fabrication-ready specifications
- Joint IP assessment

Innovation & Differentiation

Key Innovations

1. Joint HEA–HEC Optimisation

Simultaneous optimisation of:

- metallic composition,
- ceramic chemistry,
- and gradient architecture inside one AI loop.

2. Entropy-Stabilized Hybrid Interfaces

The proposal introduces entropy-compatible HEA ↔ HEO/HEC interfaces, a highly novel concept for aerospace materials engineering.

3. Physics-Informed AI

PINN-based surrogate models extrapolate beyond sparse experimental datasets.

4. Aerospace-Specific Design

The platform directly targets turbine shrouds, blade tips, combustion liners, and hot-section components relevant to Safran engine programs.

Strategic Alignment with Safran

The proposal directly addresses the Safran EXPLORE Smart Materials Challenge requirements:

Safran Requirement	HE-MCGA Response
Hybrid metal–ceramic systems	HEA ↔ HEO/HEC architectures
Engineered gradient architectures	AI-designed compositional gradients
Tunable thermo-mechanical properties	Multi-objective optimisation
AI-driven material design	End-to-end inverse design platform
In-silico performance screening	Integrated CALPHAD + FEA validation

Business Vision

The HE-MCGA platform is positioned as an AI-powered aerospace materials intelligence engine that accelerates discovery of extreme-environment hybrid architectures.

Rather than competing with fabrication facilities, the startup functions as:

- an upstream AI design platform,

- enabling rapid screening,
- optimisation,
- and digital validation before experimental synthesis.

The long-term roadmap includes:

- AI-guided additive manufacturing integration
- Autonomous materials discovery
- Digital twin development
- Hypersonic and space propulsion materials

Conclusion

The HE-MCGA platform establishes a next-generation AI-driven framework for accelerated discovery of entropy-stabilized hybrid metal–ceramic architectures capable of surviving future aerospace engine environments.

By integrating:

- high-entropy materials science,
- functionally graded architectures,
- physics-informed AI,
- thermodynamic modelling,
- and thermo-mechanical simulation,

the platform enables rapid development of materials beyond the capability of conventional aerospace alloy systems.

The proposed system positions AI not as a supplementary tool, but as the central engine driving future aerospace materials innovation.